

Matrices: Engineering Mathematics Study Notes

2. Gauss-Seidel Method

High-Level Overview

In the Jacobi method, updating variables in the $(k + 1)^{\text{th}}$ iteration relies solely on values from the previous k^{th} iteration. The **Gauss-Seidel method** introduces a crucial modification to accelerate convergence: as soon as a new value for a variable is computed, it is **immediately used** in the calculation of the subsequent variables within the same iteration cycle.

Because it uses improved components right away, this approach is also known as the **method of successive displacements**. It typically converges about twice as fast as the Jacobi method.

Step-by-Step Procedure & Iteration Formula

- Rearrange for Diagonal Dominance:** Ensure the system is diagonally dominant (reordering rows if necessary).
- Isolate Variables:** Rewrite each equation to isolate one variable on the left side.
- Apply the Successive Update Rule:** When computing values at the $(k + 1)^{\text{th}}$ iteration, substitute the newest available values:

$$x_1^{(k+1)} = \frac{1}{a_{11}} \left[b_1 - a_{12}x_2^{(k)} - a_{13}x_3^{(k)} - \cdots - a_{1n}x_n^{(k)} \right]$$

$$x_2^{(k+1)} = \frac{1}{a_{22}} \left[b_2 - a_{21}x_1^{(k+1)} - a_{23}x_3^{(k)} - \cdots - a_{2n}x_n^{(k)} \right]$$

$\vdots$

$$x_n^{(k+1)} = \frac{1}{a_{nn}} \left[b_n - a_{n1}x_1^{(k+1)} - a_{n2}x_2^{(k+1)} - \cdots - a_{n(n-1)}x_{n-1}^{(k+1)} \right]$$

- Convergence Note:** Both Jacobi and Gauss-Seidel methods converge for any choice of initial approximation $x_i^{(0)}$ if the coefficient matrix satisfies the diagonal dominance condition:

$$\sum_{j=1, j \neq i}^n \left| \frac{a_{ij}}{a_{ii}} \right| \leq 1 \quad (i = 1, 2, \dots, n)$$

Worked Examples

Example 1: Solving a System via Gauss-Seidel Method

Solve the system using the Gauss-Seidel iteration method:

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25$$

Step 1: Isolate Variables (System is Diagonally Dominant)

$$x = \frac{1}{20}(17 - y + 2z)$$

$$y = \frac{1}{20}(-18 - 3x + z)$$

$$z = \frac{1}{20}(25 - 2x + 3y)$$

Step 2: Set Initial Approximations

Let $x^{(0)} = y^{(0)} = z^{(0)} = 0$.

Step 3: Perform Iterations

- **First Iteration:**

$$x^{(1)} = \frac{1}{20}(17 - 0 + 2(0)) = 0.85$$

$$y^{(1)} = \frac{1}{20}(-18 - 3(0.85) + 0) = -1.0275$$

$$z^{(1)} = \frac{1}{20}(25 - 2(0.85) + 3(-1.0275)) = 1.019$$

- **Second Iteration (Using updated $x^{(1)}$ and $y^{(1)}$ immediately):**

$$x^{(2)} = \frac{1}{20}(17 - (-1.0275) + 2(1.019)) = 1.0025$$

$$y^{(2)} = \frac{1}{20}(-18 - 3(1.0025) + 1.019) = -0.9998$$

$$z^{(2)} = \frac{1}{20}(25 - 2(1.0025) + 3(-0.9998)) = 0.9998$$

- **Third Iteration:**

$$x^{(3)} = \frac{1}{20}(17 - (-0.9998) + 2(0.9998)) = 1.0000$$

$$y^{(3)} = \frac{1}{20}(-18 - 3(1.0000) + 0.9998) = -1.0000$$

$$z^{(3)} = \frac{1}{20}(25 - 2(1.0000) + 3(-1.0000)) = 1.0000$$

Final Answer:

$$x = 1, \quad y = -1, \quad z = 1$$

Example 2: System Requiring Row Exchange

Find the solution correct to two decimal places using the Gauss-Seidel method given the initial approximation $x_1 = 0.9, x_2 = -3.1, x_3 = 0.9$:

$$3x_1 - 6x_2 + 2x_3 = 23$$

$$-4x_1 + x_2 - x_3 = -8$$

$$x_1 - 3x_2 + 7x_3 = 17$$

Step 1: Check and Fix Diagonal Dominance

The original system is not diagonally dominant. Exchanging the first and second equations yields:

$$-4x_1 + x_2 - x_3 = -8$$

$$3x_1 - 6x_2 + 2x_3 = 23$$

$$x_1 - 3x_2 + 7x_3 = 17$$

This new system is diagonally dominant.

Step 2: Isolate Variables

$$x_1 = \frac{1}{4}(8 + x_2 - x_3)$$

$$x_2 = \frac{1}{6}(-23 + 3x_1 + 2x_3)$$

$$x_3 = \frac{1}{7}(17 - x_1 + 3x_2)$$

Step 3: Tabulate Iteration Results ($x_1^{(0)} = 0.9, x_2^{(0)} = -3.1, x_3^{(0)} = 0.9$)

Iteration (n)	x_1 (x)	x_2 (y)	x_3 (z)
1	1.0	-3.0333	0.9857
2	0.9953	-3.0071	0.9976
3	0.9988	-3.0014	0.9996

Iteration (n)	x_1 (x)	x_2 (y)	x_3 (z)
4	0.9998	-3.0002	0.9999

Final Answer:

Rounded to two decimal places:

$x_1 = 0.99, \quad x_2 = -3.00, \quad x_3 = 0.99$